DIGITAL FORENSICS INVESTIGATION REPORT

Lockdown Lab – Network, Memory & Malware Forensics

Case Name	Lockdown Lab Investigation
Case Number	2026-011
Examiner	Tej Chaute
Date	07/04/2026
Platform	CyberDefenders
Evidence File	capture.pcapng, memdump.mem, updatenow.exe
SHA256 Hash	cd93ae35b3a4989232dae009bdfb93cbf148ccbfe5087b96c9ad823034ca0bfb, 94ffad6a879395ec1ce438dfc1347e825200767c07d3e2a6167a34fb764ce64f, c25a6673a24d169de1bb399d226c12cdc666e0fa534149fc9fa7896ee61d406f
Classification	Training

1. Case Summary

The investigation focused on identifying and reconstructing a cyber intrusion involving an IIS web server. Analysis of network traffic and memory artifacts revealed that an external attacker performed reconnaissance, gained access via SMB, deployed a web shell, established a reverse shell, and maintained persistence using a startup-based implant. The attack was ultimately attributed to the Agent Tesla malware family, confirming a full compromise lifecycle.

2. Evidence Examined

Item	Filename	Hash (SHA256)	File Size
001	capture.pcapng	cd93ae35b3a4989232dae009bdfb93cbf148ccbfe5087b96c9ad823034ca0bfb	3.7 MB
002	memdump.mem	94ffad6a879395ec1ce438dfc1347e825200767c07d3e2a6167a34fb764ce64f	4.5 GB
003	updatenow.exe	c25a6673a24d169de1bb399d226c12cdc666e0fa534149fc9fa7896ee61d406f	588 KB

3. Tools Used

Tool	Version	Purpose
Wireshark	4.6.4	Network traffic analysis — TCP stream reconstruction, SMB analysis, C2 identification
Volatility 3	2.28.1	Memory forensics — process analysis, network connections, persistence detection
Strings	2.46	Static string extraction from updatenow.exe for malware analysis
VirusTotal	2.5.8	Threat intelligence — malware family attribution via hash correlation

4. Key Findings

Each finding below is supported by at least one artifact. Path, timestamp, and source tool are documented for every item.

4.1 System Information

Field	Value
OS	Windows 10 (Build 17763)

Field	Value
Timezone	UTC
System Time	2024-09-10 06:14:13 UTC
Kernel Base	0xf80079213000
Architecture	x64

4.2 Artifact Table

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
1	Network Traffic (PCAP)	TCP SYN Scan (10.0.2.4 → 10.0.2.15)	2024-09-10 05:44:28	The attacker initiated reconnaissance by performing a high-volume TCP SYN scan against the IIS server, identifying open ports and mapping available services for potential exploitation.
2	Network Traffic (PCAP)	Continued TCP Probing	2024-09-10 05:44:30+	Sustained scanning activity confirms automated or tool-based reconnaissance, indicating deliberate service discovery rather than accidental traffic.
3	SMB Access (IPC$)	\\10.0.2.15\IPC$	2024-09-10 05:47:11	The attacker successfully interacted with SMB services and accessed the IPC$ administrative share, confirming network-level access and authentication capability on the target system.
4	SMB Access (Documents)	\\10.0.2.15\Documents	2024-09-10 05:47:33	Access to a user-accessible share demonstrates enumeration of internal resources, indicating progression from external reconnaissance to internal exploration.
5	SMB File Upload	PCAP — SMB write shell.aspx	2024-09-10 05:48:52	The attacker uploaded a malicious ASP.NET web shell to the server, enabling remote code execution via the IIS web

#	Artifact Type	Location / Path	Timestamp (UTC)	What It Proves
				service. This marks the initial compromise of the system.
6	Reverse Shell (C2)	PCAP — outbound TCP Port 4443	2024-09-10 06:08:23	Compromised server initiated outbound connection to attacker on port 4443 — confirms active reverse shell granting interactive remote command execution; outbound direction bypasses inbound firewall rules
7	Process Execution	w3wp.exe → updatenow.exe	2024-09-10 06:08:23	IIS worker process (w3wp.exe) spawned updatenow.exe — process spawned by a web server worker outside normal operations confirms execution of attacker payload delivered via the web shell
8	Persistence (Startup)	C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe	—	Malware placed in the Windows Startup folder — ensures automatic execution on every system reboot, providing persistent access independent of the web shell or active session
9	Malware Analysis	strings analysis updatenow.exe	—	updatenow.exe identified as UPX-packed — deliberate executable packing used to reduce binary size and evade signature-based detection; packing must be removed before deeper static analysis
10	Threat Intelligence	VirusTotal (SHA256 hash)	—	Hash correlation via VirusTotal attributes updatenow.exe to the Agent Tesla malware family — a known commodity RAT specialising in credential theft from browsers, email clients, and FTP applications via keylogging and screenshot capture

5. Event Timeline

Reconstructed chronological sequence of events based on artifact timestamps. All times in UTC.

Timestamp (UTC)	Artifact Source	Event Description
2024-09-10 05:44:28	PCAP	The attacker (10.0.2.4) initiated reconnaissance by sending a burst of TCP SYN packets to the IIS server (10.0.2.15), attempting to identify open ports and exposed services.
2024-09-10 05:44:30	PCAP	Continued scanning activity confirmed systematic service enumeration, indicating the attacker was mapping the attack surface before exploitation.
2024-09-10 05:47:11	SMB Traffic	The attacker accessed the IPC$ share, confirming successful SMB communication and the ability to interact with administrative services on the target system.
2024-09-10 05:47:33	SMB Traffic	The attacker accessed the Documents share, demonstrating internal enumeration and potential targeting of sensitive files or writable locations.
2024-09-10 05:48:52	SMB Write	A malicious web shell (shell.aspx) was uploaded to the IIS server, enabling remote execution of commands via HTTP requests. This marks the point of initial system compromise.
2024-09-10 06:08:23	Network Traffic	The compromised system initiated an outbound connection to the attacker on port 4443, establishing a reverse shell and granting interactive control to the attacker.
2024-09-10 06:08:23	Memory (pslist)	The IIS worker process (w3wp.exe) spawned updatenow.exe, indicating execution of a secondary payload delivered via the web shell.
2024-09-10 06:08:23+	Persistence Artifact	The attacker placed updatenow.exe in the Startup folder, ensuring persistence across reboots and maintaining long-term access to the system.
—	strings / VirusTotal	updatenow.exe identified as UPX-packed Agent Tesla RAT — credential-stealing malware confirmed via hash attribution

6. Conclusion

The investigation confirms that the IIS web server (10.0.2.15) was fully compromised through a structured, multi-stage attack originating from 10.0.2.4.

Reconnaissance and Initial Access: The attacker began with a TCP SYN scan to identify open services (Artifact #1), then pivoted to SMB to authenticate and enumerate internal shares (Artifacts #2–3). The IPC$ and Documents shares were accessed before shell.aspx was written to the server via SMB (Artifact #4) — the upload itself required no web application vulnerability; the attacker wrote directly to a location accessible by IIS.

Execution and Command & Control: The web shell enabled remote command execution via HTTP, through which updatenow.exe was delivered and run by the IIS worker process (Artifact #6). The compromised server subsequently established an outbound reverse shell to the attacker on port 4443 (Artifact #5), granting interactive control while bypassing inbound firewall restrictions.

Persistence and Attribution: updatenow.exe was placed in the Windows Startup folder (Artifact #7), ensuring execution on every reboot independent of the active session or web shell. String extraction identified UPX packing (Artifact #8), and VirusTotal hash attribution confirmed the sample as Agent Tesla (Artifact #9) — a commodity RAT capable of keylogging, screenshot capture, and credential theft from browsers, email clients, and FTP applications.

The attack chain — SYN scan, SMB access, web shell upload, reverse shell, secondary payload, and startup persistence — represents a complete intrusion lifecycle with all five stages: initial access, execution, persistence, command and control, and collection.

7. Examiner Notes

This investigation was conducted retrospectively as part of a structured DFIR learning path. It is the first case in this portfolio to combine three distinct evidence types — network capture, memory dump, and a standalone malware sample — requiring correlation across all three sources to reconstruct the full attack chain.

Key methodological observations from this case:

- Initial attempts to identify the C2 domain through DNS traffic analysis and string extraction from the memory dump produced significant noise and false positives. The correct attribution came from pivoting to VirusTotal using the updatenow.exe SHA256 hash, which provided reliable family identification and relational analysis. This reinforces that local evidence is not always sufficient — threat intelligence platforms fill gaps that artifact analysis alone cannot.

- The attack did not require a web application vulnerability for initial compromise. The attacker wrote shell.aspx directly via SMB using credentials or an authenticated session — no injection, no exploit. This is a reminder that network-level access controls (SMB exposure, credential hygiene) are as important as application-layer security.
- w3wp.exe spawning updatenow.exe is a high-fidelity indicator of web shell execution. IIS worker processes should never spawn arbitrary executables — any process tree showing w3wp.exe as a parent of a non-IIS binary warrants immediate investigation. This is a reliable detection rule for web shell post-exploitation.
- UPX packing does not require specialised tools to identify — the string 'UPX' appears in the binary header and is visible through basic string extraction. However, packed binaries must be unpacked before meaningful static analysis (import tables, strings, code structure) can be performed. VirusTotal's sandbox unpacks and analyses the sample automatically, which is why hash submission was more productive than local string extraction in this case.
- Startup folder persistence is one of the simplest and most reliable persistence mechanisms on Windows. C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\ is world-writable by default in many configurations, meaning any user-level process can drop a file there without requiring elevated privileges. Detection is straightforward — monitoring new file writes to Startup folders is a low-cost, high-signal detection opportunity.

If this were a real-world investigation, recommended next steps would include: immediately removing shell.aspx from the IIS server and blocking SMB access from the attacker IP; rotating all credentials accessible on the compromised system; extracting credentials targeted by Agent Tesla from browser profiles, email clients, and FTP configurations; reviewing the system for additional persistence mechanisms (registry Run keys, scheduled tasks); and investigating whether the compromised system was used for lateral movement to other internal hosts.